Abstract for the Tsukuba Science Edge 2026 [English form]

Your research stage is : □**Planning** □**Preliminary** ■**Result** 【Required: Pick only one】

Area Check Most appropriate one【Required】

□Physics □Chemistry ■ Biology & Medical □Earth Sciences □Mathematics＆Information

Participant's Information **【Published as are written here in English】**

【School / Organization Name】Taipei Municipal Lishan High School

【Representative's Name (top author)】LEE CHIH YUN

【Member's Name】HSIN JIA CHEN

Supervising Teacher(s) / Person(s) 【Responsible for content guarantee and inquiry response】

【Name】KUO CHIUNG HUA；LIN SHIANN SHENG

Abstract of the Presentation

【Title】 Biomechanical Analysis of Locomotion and Morphological Adaptations in Snakes from Distinct Habitats

【 標題 】不同棲息地蛇類的運動生物力學分析與形態適應

【Outline and its First Inspiration】

During a campus observation, we encountered a wild *Boiga kraepelini*. We were surprised to find that this species, within the forest canopy, can coil their tails on to our fingers and extend to survey its surroundings. This observation prompted our investigation about what specific anatomical structures grant them an advantage in climbing.

To investigate the physical trade-offs in locomotion strategies between arboreal and terrestrial snakes, we initiated this study. This experiment aims to determine the relationship between body anatomy and locomotion limits. We selected four representative species for comparison: the arboreal *Boiga kraepelini*, the terrestrial *Heterodon nasicus*, *Pantherophis guttatus*, which appears in many habitats, and *Boa constrictor*, which is arboreal but lives mostly on the ground upon reaching maturity. (Figure 1)

The experiment is divided into three parts:

Body dimensions: Quantifying and comparing the head/body/tail proportions of each species. Including the length and width of each body part. (Figure 2,3,4)

Horizontal reaching test: Simulating suspended movement between branches to test the snakes' cantilever limit in the absence of support, while exploring the impact of gravitational torque on different body types.

Vertical extension test: Utilizing apparatuses of varying heights to test the snakes' ability to overcome gravity during upward movement, verifying whether arboreal snakes utilize their ventral scales or tails as critical anchoring points. (Figure 5,6)

By analyzing this locomotion data, we hope to not only unravel the evolutionary mysteries of snakes but also provide a biomechanical foundation for future ecological engineering. Such as designing ecorridors for snakes to reduce the impact of human infrastructure on their survival.

在一次校園觀察中，我們遇到了一條野生的黑眉錦蛇（Boiga kraepelini）。我們驚訝地發現，這種蛇在森林樹冠層中，能夠將尾巴纏繞在我們的手指上，並伸展身體偵測周圍環境。這項觀察促使我們展開研究，探討究竟是哪些特定的解剖結構賦予了它們攀爬的優勢。

為了研究樹棲和陸棲蛇類在移動策略上的生理權衡，我們啟動了這項研究。本實驗旨在確定身體解剖結構與移動極限之間的關係。我們選擇了四種具有代表性的蛇類進行比較：樹棲的黑眉錦蛇（Boiga kraepelini）、陸棲的西部豬鼻蛇（Heterodon nasicus）、出現在多種棲息地的玉米錦蛇（Pantherophis guttatus），以及雖為樹棲但成年後多生活在地面的紅尾蚺（Boa constrictor）（圖 1）。

實驗分為三個部分：

體型測量：量化並比較每個物種的頭部/身體/尾部比例，包括每個身體部位的長度和寬度（圖 2、3、4）。

水平伸展測試：模擬樹枝間的懸空移動，測試蛇在無支撐狀態下的懸臂極限，同時探究重力扭矩對不同體型的影響。

垂直伸展測試：利用不同高度的裝置，測試蛇在向上移動時克服重力的能力，驗證樹棲蛇是否利用其腹鱗或尾巴作為關鍵的固定點（圖 5、6）。

通過分析這些移動數據，我們不僅希望揭開蛇類的演化之謎，還能為未來的生態工程提供生物力學基礎，例如設計蛇類的生態廊道，以減少人類基礎設施對它們生存的影響。

【Novelty of Your Goal and its Original Value】

Existing literature on snake locomotion focuses on qualitative patterns such as serpentine crawling and sidewinding. However, there is a lack of quantitative data regarding the biomechanical limits of snakes when challenging gravity. Our study is novel in that it shifts the focus to how far they can extend before physical failure, specifically defining the cantilever limits in both vertical and horizontal planes.

The original value of this research lies in correlating these physical limits with specific morphological

adaptations. By comparing the four snakes with different body structures and habitats, we quantified how form defines function.

Our experiments revealed a potential universal biomechanical constraint, showing that despite vast differences in size and habitat, the horizontal reaching limit for snakes generally falls between 36% and 43% of their total body length. Furthermore, we established that a higher tail-to-body ratio is a critical predictor for vertical climbing success, providing new quantitative metrics for understanding snake evolution and habitat adaptation. (Table 6-8)

現有關於蛇類移動的文獻主要著重於定性模式，例如蛇形爬行和側行。然而，在挑戰重力時，有關蛇類生物力學極限的定量數據相當缺乏。我們的研究具有創新性，因為它將焦點轉移到蛇類在物理崩潰前能延伸多遠，特別是定義了垂直和水平平面上的懸臂極限。

本研究的獨特價值在於將這些物理極限與特定的形態適應相關聯。通過比較四種具有不同身體結構和棲息地的蛇，我們量化了形態如何決定功能。

我們的實驗揭示了一個潛在的普遍生物力學限制，結果顯示儘管體型和棲息地存在巨大差異，蛇類的水平伸展極限通常介於其總體長的 36%至 43%之間。此外，我們確定較高的尾身比是垂直攀爬成功與否的關鍵預測因素，這為理解蛇類的演化和棲息地適應提供了新的定量指標（表 6-8）。

【Creativity of Your Research Plan】

Rather than relying only on observation, we designed the experiment with height differentials to bring out the snakes' locomotor limits. The experiment consisted of a “vertical extension test” and a “horizontal reaching test,” (Figure 4,5) combined with video analysis to observe muscle control and failure patterns (e.g., pulling back due to insufficient tension) during extreme extension.

At the beginning of the experiment, we initially used branches to encourage the snakes to stretch upward. Later, we decided to use containers of varying heights to test whether the snakes could climb out. This adjustment not only increased the convenience of the experiment but also reduced human interference, allowing for more reliable conclusions.

Additionally, using containers provided a clearer demonstration of how humans can safely husbandry and relocation" 。 snakes back into the wild, aligning with the United Nations Sustainable Development Goal 15 (SDG15).

During the experiment, we continuously monitored the snakes' conditions to ensure they were motivated and willing to participate. At the same time, we maintained a safe distance between the snakes to avoid excessive interference.

我們不僅僅依賴觀察，還設計了具有高度差異的實驗，以揭示蛇的移動極限。該實驗包括「垂直延伸測試」和「水平伸展測試」(圖 4、5)，並結合視頻分析，觀察蛇在極度伸展過程中的肌肉控制和失敗模式（例如，因張力不足而後退）。

實驗開始時，我們最初使用樹枝鼓勵蛇向上伸展。後來，我們決定使用不同高度的容器來測試蛇是否能夠爬出來。這項調整不僅提高了實驗的便利性，還減少了人為干擾，從而得出更可靠的結論。

此外，使用容器更清晰地展示了人類如何安全地飼養蛇並將其重新放歸野外，這與聯合國可持續發展目標 15（SDG15）相符。

在實驗過程中，我們持續監測蛇的狀況，確保它們有動力且願意參與。與此同時，我們在蛇之間保持安全距離，以避免過度干擾。

【Feasibility and its Prospect, based on Your Result and Interpretation】

1. Horizontal reaching: Our experiments successfully quantified the cantilever limits of snakes. Despite significant size differences (47 cm to 150 cm) and habitat differences, we identified a universal horizontal reaching limit occurring between 42.6% and 46.7% of the total body length across snake species. This consistency confirms our method and indicates a fundamental physical limit in limbless vertebrates under gravity. (Table 6-8 and Graph 4)

2. Interpretation of Morphological Trade-offs: By correlating morphological data with locomotor performance, we interpreted the physical costs of evolutionary adaptations:

･The fossorial *Heterodon nasicus*, possessing the largest head ratio (6.4%), showed the poorest stability. Its success rate dropped significantly at just 31.9% extension, demonstrating how a heavy head specialized for digging increases anterior torque and horizontal reaching ability. (Table 1 and Graph 4)

･Long Tail Advantage: Conversely, the arboreal *Boiga kraepelini*, with larger tail-to-body ratio (27.9%), demonstrated superior performance. It achieved a vertical extension of 55.8% (48cm) of its body length. This interprets the tail's critical biomechanical function as a "posterior anchor" and counterweight in arboreal locomotion. (Table 1 and 5)

3. Differentiation of Locomotor Strategies: Comparing *Boa constrictor* and *Boiga kraepelini* reveals two distinct strategies to overcome gravity. The *Boa constrictor*, despite having the shortest tail (9.3%), utilized its highest body ratio (86.4%) and muscle mass to achieve a 43% horizontal limit. This proves that snakes have evolved different strategies, using strong body muscles (*Boa constrictor*) and using long tails (*Boiga kraepelini*). Yet both are ultimately bound by the ~43% limit. (Graph 1,2,3)

4. Restriction of friction and base area: During our vertical extension test, we found out that when the *Boa constrictor* with the longest body length was placed in a plastic, small base area container, it was hard for it to

extend and climb out. However, when we switch to a box with more friction and a larger base, the *Boa constrictor* easily climbs out. These results demonstrate that the snake's vertical extension ability is highly dependent on a container's friction and base area. (Table 4) Note that *Boiga kraepelini* consistently reached the greatest vertical extension relative to body length, with values approaching 50% of total length.(Table 5) Overall, the results indicate that while absolute climbing height varies among species, vertical extension is constrained within a comparable proportional range, suggesting a shared biomechanical limitation in upward climbing among snakes.

1. 水平伸展：我們的實驗成功量化了蛇的懸臂極限。儘管存在顯著的體型差異（47 公分至 150 公分）和棲息地差異，但我們發現所有蛇類的水平伸展極限普遍在其身體總長的 42.6%至 46.7%之間。這種一致性驗證了我們的方法，並表明無肢脊椎動物在重力作用下存在基本的物理極限。（表 6-8 和圖 4）
2. 形態學權衡的解釋：通過將形態學數據與運動表現相關聯，我們解釋了進化適應的物理成本：
・ 穴居的 Heterodon nasicus 具有最大的頭部比例（6.4%），但其穩定性最差。當伸展僅達 31.9%時，其成功率顯著下降，這表明專門用於挖掘的沉重頭部會增加前部扭矩，從而影響水平伸展能力。（表 1 和圖 4）
・ 長尾優勢：相反，樹棲的 Boiga kraepelini 具有較大的尾體比例（27.9%），表現更優。它實現了體長 55.8%（48 公分）的垂直伸展。這解釋了在樹棲運動中，尾巴作為「後部錨點」和平衡重的關鍵生物力學功能。（表 1 和表 5）
3. 運動策略的差異：對比紅尾蚺（Boa constrictor）和 Boiga kraepelini，發現了兩種截然不同的克服重力的策略。紅尾蚺儘管尾巴最短（9.3%），但利用其最高的體幹比例（86.4%）和肌肉質量達到了 43%的水平極限。這證明蛇類已演化出不同的策略，諸如利用強壯的體幹肌肉（紅尾蚺）和利用長尾巴（Boiga kraepelini）。然而，最終兩者都受制於約 43%的極限。（圖 1、2、3）
4. 摩擦力和基底面積的限制：在我們的垂直伸展測試中，我們發現當體長最長的紅尾蚺被置於塑料製、基底面積小的容器中時，它很難伸展並爬出。然而，當我們換成摩擦力更大、基底面積更大的箱子時，紅尾蚺輕易地爬了出來。這些結果表明，蛇的垂直伸展能力高度依賴於容器的摩擦力和基底面積。（表 4）值得注意的是，相對於體長而言，Boiga kraepelini 的垂直伸展始終是最大的，其數值接近體長的 50%。（表 5）總體而言，結果表明儘管不同物種的絕對攀爬高度存在差異，但垂直伸展都受限於一個相近的比例範圍，這暗示了蛇類在向上攀爬時存在共同的生物力學限制。

【Future Potential and Your Plan, if any】

1. Scientifically Informed Coexistence (SDG 15): Our findings provide a basis for human-snake coexistence. By establishing that the vertical extending limit for snakes is approximately 43% of their body length, we can design habitat barriers. For instance, a physical barrier that is 50% of the local snake's size would effectively prevent snakes from entering residential areas without harming them. Furthermore, understanding that arboreal species are specialized for climbing, we propose designing canopy bridges rather than underpasses to reduce roadkill, directly addressing SDG 15 (Life on Land).
2. Rescue Robotics: Current snake robots struggle with gap crossing. By mimicking the strategy of using a high-tail-ratio structure as a counterbalance and anchor, we aim to inspire robots that can navigate discontinuous terrains by performing stable cantilevered extensions to reach victims.
3. Future Research: We plan to expand our dataset to include more diverse species to verify if the "43% limit" is a universal biomechanical constant. Additionally, we aim to collaborate with biophysicists to study how scale micro-structures influence friction during vertical climbing, translating these biological traits into safer coexistence devices and advanced engineering solutions.

1. 基於科學的共存（永續發展目標 15）：我們的研究結果為人類與蛇類的共存提供了依據。透過確定蛇類的垂直延伸極限約為其身體長度的 43%，我們可以設計棲息地屏障。例如，一個高度為當地蛇類體長 50%的物理屏障，就能有效防止蛇類進入住宅區，同時不會傷害它們。此外，瞭解到樹棲物種擅長攀爬，我們建議設計樹冠橋而非地下通道，以減少路殺現象，直接對應永續發展目標 15（陸地生命）。
2. 救援機器人：現有的蛇形機器人在跨越間隙方面存在困難。透過模仿使用高尾部比例結構作為平衡物和錨點的策略，我們旨在研發能夠通過穩定的懸臂延伸來穿越不連續地形，從而抵達受困者的機器人。
3. 未來研究：我們計劃擴大資料集，納入更多樣化的物種，以驗證「43%極限」是否為一個普遍的生物力學常數。此外，我們旨在與生物物理學家合作，研究鱗片微結構如何影響垂直攀爬時的摩擦力，將這些生物特徵轉化為更安全的共存裝置和先進的工程解決方案。

【References 】

1. Sheehy, Coleman M., et al. (2016). The evolution of tail length in snakes associated with different gravitational environments.
2. Greg Byrnes, Bruce C Jayne (2012). The effects of three-dimensional gap orientation on bridging performance and behavior of brown tree snakes (*Boiga irregularis*).
3. Bruce C Jayne, et al. (2015). Why arboreal snakes should not be cylindrical: body shape, incline and surface roughness have interactive effects on locomotion.

※You may fill this form **up to 2 printed pages** plus additional materials as below, but the latte will not be printed publication.
※**Your method and results** are required for reviewing so you must attach additional tables, charts, photos, etc., as many as needed. Please put respective number, title and caption. Also in the References list, all necessary references are numbered and referred in the text.
※For each description, please refer to the **Start-Up Check List**.

Figure 1: Basic information of experiment snakes

Figure 2,3,4:

Figure 2,3,4 shows the image of measuring snake's body dimension
The following is a description of our steps and method:

Total length of snake
1.The tip of the tail of experiment snake was aligned with the knotted end of a rope
2.Place the rope along the length of the snake's body.
3.Mark the position corresponding to the tip of the head on the rope. The distance between the knot and the marked point was recorded as the total body length.
For each region, repeat the process 3 times and record.

Head/body/tail measurements
1.The tip of the head was aligned with the zero mark of a ruler.
2.The following three anatomical landmarks were sequentially measured, recorded, and photographed:
(a) the posterior margin of the head (junction between head and neck)
(b) the lowest point of the neck where body width no longer changes
(c) the base of the tail (at the cloacal region).
For each region, repeat the process 3 times and record.

Width measurement

1.Use a digital caliper to measure the widths of the head, neck, body, and tail. For each region, measurements were taken at the visually widest point and repeated three times.

All repeated measurements were recorded in a data table, and mean values were calculated. These means were used as the final values for total body length and for the lengths and widths of the head, neck, body, and tail.

圖 2、3、4 顯示了測量蛇體尺寸的圖像

以下是我們的步驟和方法說明：

蛇的總長度

1. 將實驗蛇的尾尖與繩子的打結端對齊
2. 將繩子沿著蛇的身體長度放置。
3. 在繩子上標出與頭尖相對應的位置。結與標記點之間的距離記錄為總體長。

每個區域重複此過程 3 次並記錄。

頭部/身體/尾部測量

1. 將頭尖與尺子的零刻度對齊。
2. 依次對以下三個解剖學標誌進行測量、記錄和拍攝：

(a) 頭部後緣（頭部與頸部的連接處）

(b) 頸部寬度不再變化的最低點

(c) 尾部基部（泄殖腔區域）。

每個區域重複此過程 3 次並記錄。

寬度測量

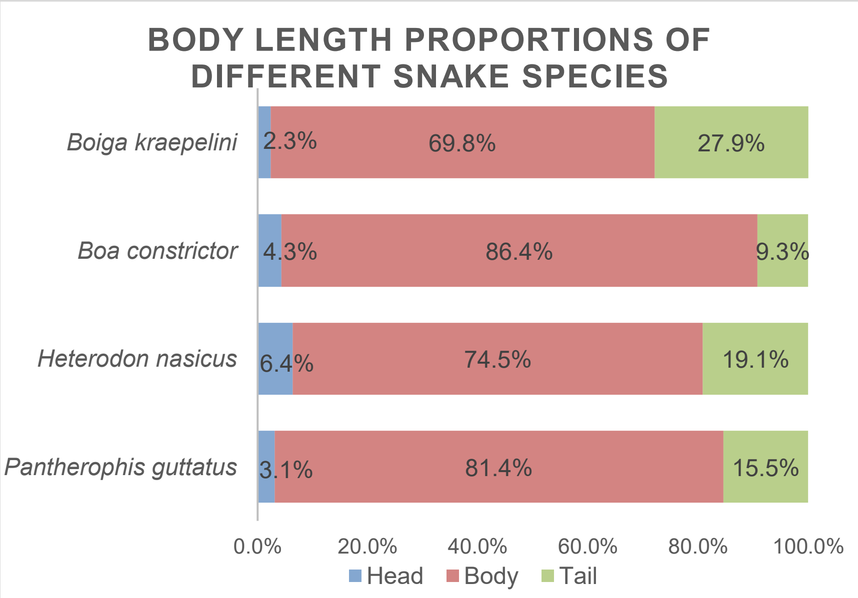
1. 使用數字卡尺測量頭部、頸部、身體和尾部的寬度。每個區域在視覺上最寬的點進行測量，並重複三次。

所有重複測量均記錄在數據表中，並計算平均值。這些平均值用作總體長以及頭部、頸部、身體和尾部的長度與寬度的最終數值。

Abstract for the Tsukuba Science Edge 2026 [English form]

Table 1: Body dimensions of each experiment snake

	Pantherophis guttatus	*Heterodon nasicus*	*Boa constrictor*	*Boiga kraepelini*
Total(cm)	97	47	150.3	86
Head	3.1%	6.4%	4.3%	2.3%
Body	81.4%	74.5%	86.4%	69.8%
Tail	15.5%	19.1%	9.3%	27.9%



Graph 1: Body length proportions of each experiment snake

Table 1 and Graph 1 shows a highly significant trend that can be observed: *Boiga kraepelini* exhibited the smallest head ratio (2.3%) and the largest tail ratio (27.9%). From a biomechanical perspective, a smaller head reduces tip load during horizontal reaching, while a longer tail provides stronger posterior anchoring effects.

In contrast, although *Boa constrictor* possesses a much larger body size, its tail accounts for only 9.3% of total body length. This suggests that during locomotion, *Boa constrictor* may rely more heavily on forces generated by powerful body muscle to maintain grip and generate upward force, rather than on tail-based anchoring mechanisms.

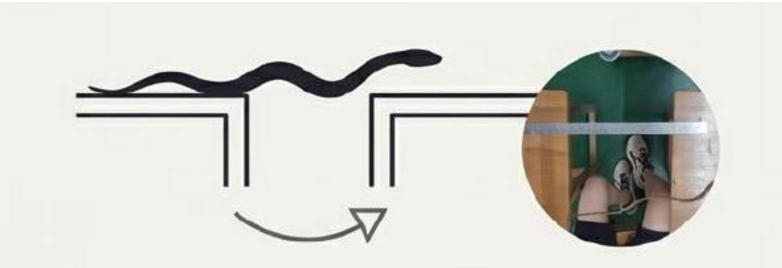
Heterodon nasicus, in contrast, has a relatively large head proportion(6.4%), which is advantageous for digging in a fossorial habitat but is associated with reduced locomotor performance.

圖 1：每條實驗蛇的體長比例

表 1 和圖 1 顯示出一個極為顯著的趨勢：大頭蛇（Boiga kraepelini）呈現出最小的頭部比例（2.3%）和最大的尾部比例（27.9%）。從生物力學的角度來看，較小的頭部會減少水平伸展時的末端負荷，而較長的尾部則能提供更強的後部固定效果。

與此相反，雖然紅尾蚺（Boa constrictor）的體型大得多，但其尾部僅占總體長的 9.3%。這表明在移動過程中，紅尾蚺可能更依賴強壯的身體肌肉產生的力量來維持抓握並產生向上的力，而非基於尾部的固定機制。

相反地，北美豬鼻蛇（Heterodon nasicus）的頭部比例相對較大（6.4%），這有利於其在穴居環境中挖掘，但與此同時也會降低其運動性能。



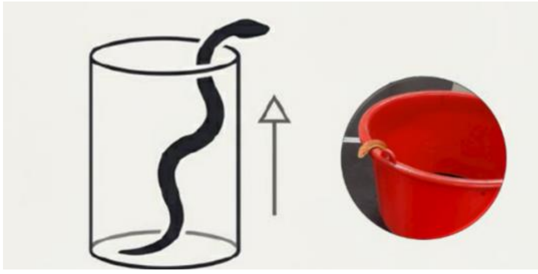


Figure 5,6: Horizontal reaching/ vertical extension image
(Generated using AI-assisted tools to create clearer image of experiment process)

The following is a description of our steps and method:

Figure 5: Horizontal reaching test
1.Place two desks of equal height parallel to each other.
2.The distance between the desk surfaces was adjusted to the target length using a ruler.
3.The experimental snake was encouraged to move forward using food or a branch, success was defined as the snake reached the target desk.
4.Each distance was tested in three trials, total number of successful trials was recorded. If less than two successful trials were observed, the distance was reduced.
5.The distance between desks was systematically increased or decreased until the maximum horizontal reaching distance was determined.
圖 5：水平伸展測試
1. 放置兩張高度相同的桌子，使其相互平行。
2. 用尺子將桌面之間的距離調整到目標長度。
3. 用食物或樹枝鼓勵實驗蛇向前移動，成功的定義是蛇到達了目標桌子。
4. 每個距離進行三次試驗，記錄成功試驗的總次數。如果觀察到的成功試驗少於兩次，則減小距離。
5. 系統地增加或減小桌子之間的距離，直到確定最大水平伸展距離。

Figure 6: Vertical extension test:

1.Prepare several plastic containers with different internal heights, measure and record their internal heights.
2. The experimental snake was first placed into the container with the lowest height, timing was initiated. If the snake climbed out of the container within 5 minutes, the trial was recorded as a success and the time required (in seconds) was documented. If the snake failed to exit the container within 5 minutes, the trial was recorded as a failure.
3.Repeat experiment 3 times for each container height. Record the number of successful trials and the time used of successful attempts. If the snake succeeded in two or more trials at a given height, the test proceeded to a container with a greater height.
4.This procedure was repeated until the snake exhibited only one or zero successful trials at a given height, which was defined as the maximum vertical extension height.

1. 準備數個內部高度不同的塑膠容器，測量並記錄它們的內部高度。

2. 首先將實驗用蛇放入高度最低的容器中，開始計時。如果蛇在 5 分鐘內爬出容器，則該次試驗記為成功，並記錄所需時間（以秒為單位）。如果蛇在 5 分鐘內未能逃出容器，則該次試驗記為失敗。

3. 每個容器高度重複實驗 3 次。記錄成功試驗的次數以及成功嘗試所花費的時間。如果蛇在某一高度的試驗中成功兩次或更多次，則測試將使用高度更高的容器繼續進行。

4. 重複此步驟，直到蛇在某一高度的試驗中僅有一次成功或零次成功為止，此高度即被定義為最大垂直伸展高度。

Table 2,3,4,5: Vertical extension result of 4 species

Heterodon nasicus body length：47cm				
Height (cm)	17	20	15(box)	23(box)
Success Rate (%)	100	0	100	0
Avg. Time (s)	112	0	64	0

Pantherophis guttatus body length：97cm				
Height (cm)	20	25	31	32
Success Rate (%)	100	66.7	33.3	66.7
Avg. Time (s)	115	76	16	145

Boa constrictor body length：150cm					
Height (cm)	20	32	46.5	37(box)	57(box)
Success Rate (%)	0	66.7	0	100	100
Avg. Time (s)	0	116	0	8	57

Boiga kraepelini body length：86cm			
	1st	2nd	Avg.
Height (cm)	35	48	41.5
Height / total body length(%)	40.7	55.8	48.3

Tables show the recorded success rates and average times at various heights:

Heterodon nasicus: This species had a 100% success rate at 17 cm. Its vertical limit is approximately 32% to 42% of its total body length (47 cm). Its large head and heavy build make it prone to losing balance without vertical support.

Pantherophis guttatus: Showed better flexibility, with a 100% success rate at 20 cm, dropping to 33.3% at 31 cm. This linear decay confirms the struggle between muscle strength and gravitational torque.

Boa constrictor: Performed poorly in small, smooth containers (0% success at 20 cm). However, when provided with a box with larger base and higher friction, it reached a 100% success rate at 57 cm. This indicates that the vertical ability of large snakes is highly dependent on contact surface area and friction.

Boiga kraepelini : Demonstrated overwhelming superiority, reaching an average vertical extension of 48.3% of its body length, with a maximum record of 55.8%. This proves the efficiency of arboreal specializations (long tail and lateral compression) in resisting gravity.

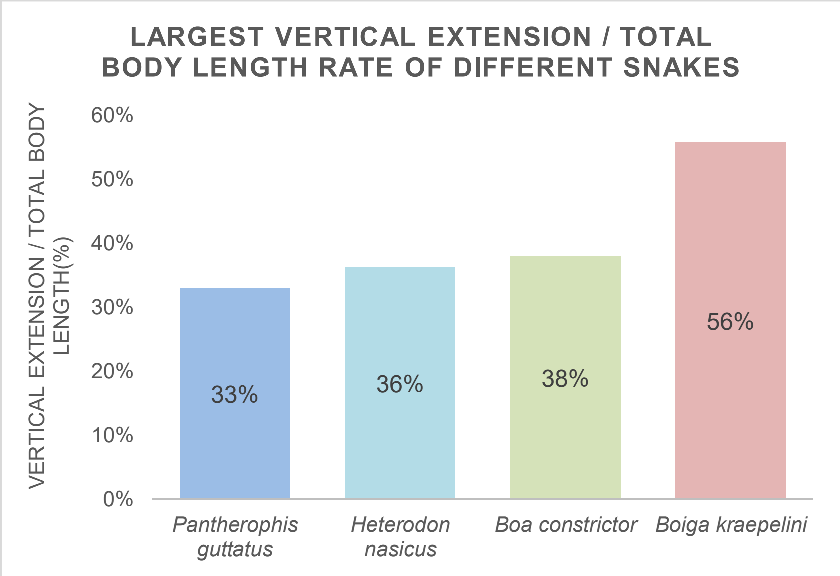
表格顯示了在不同高度下記錄的成功率和平均時間：

西部豬鼻蛇（Heterodon nasicus）：該物種在 17 厘米高度的成功率為 100%。其垂直極限約為其身體總長度（47 厘米）的 32%至 42%。其較大的頭部和厚重的體型使其在沒有垂直支撐的情況下容易失去平衡。

玉米錦蛇（Pantherophis guttatus）：表現出更好的靈活性，在 20 厘米高度的成功率為 100%，在 31 厘米高度時降至 33.3%。這種線性下降證實了肌肉力量與重力扭矩之間的較量。

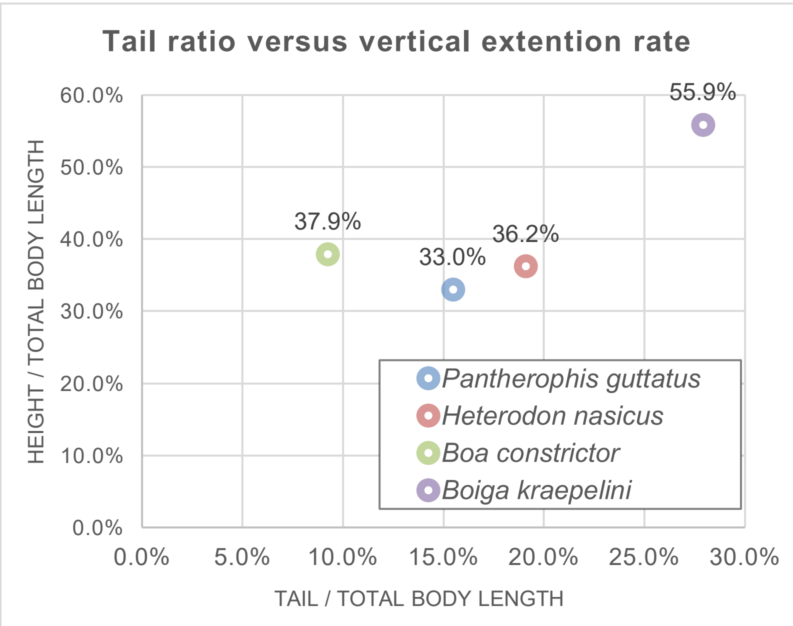
紅尾蚺（Boa constrictor）：在小型光滑容器中表現不佳（20 厘米高度的成功率為 0%）。然而，當提供一個底座更大、摩擦力更高的盒子時，它在 57 厘米高度達到了 100%的成功率。這表明大型蛇類的垂直能力高度依賴於接觸表面積和摩擦力。

大頭蛇（Boiga kraepelini）：展現出壓倒性的優勢，其平均垂直伸展長度達到其身體長度的 48.3%，最高記錄為 55.8%。這證明了樹棲特化（長尾巴和側向壓縮）在抵抗重力方面的有效性。



Graph 2: Largest vertical extension/ body length rate bar graph

Graph 2 summarizes the maximum vertical extension relative to body length for each species which clearly shows that *Boiga kraepelini* outperformed every other snake.



Graph 3: Scatter plot of tail ratio versus vertical extension rate

Graph 3 shows a clear positive correlation: snakes with a bigger tail ratio performs better in vertical extension tests. Mechanically, this is "posterior torque compensation." The tail acts as a counterweight and a friction anchor, helping balance the body and prevent forward tipping. For *Boiga kraepelini*, its 27.9% tail length enables it to extend up to 55.9% of its body while maintaining balance against gravity.

圖 3 顯示了明顯的正相關性：尾部比例較大的蛇在垂直伸展測試中表現更好。從力學角度來看，這就是「後部扭矩補償」。尾部充當配重和摩擦錨，幫助平衡身體並防止向前傾倒。對於紫棕游蛇（Boiga kraepelini）來說，其 27.9%的尾部長度使其能夠伸展至體長的 55.9%，同時保持對重力的平衡。

Table 6,7,8: Horizontal reaching result of 3 species

Heterodon nasicus body length：47cm				
Length (cm)	5	10	15	20
Success Rate (%)	100	100	66.7	0
$\frac{\text{Max reaching distance}}{\text{Total body length}}$ (%)	31.9			

Boa constrictor body length：150cm				
Length (cm)	50	60	65	70
Success Rate (%)	100	66.7	66.7	0
$\frac{\text{Max reaching distance}}{\text{Total body length}}$ (%)	43			

Boiga kraepelini body length：86cm				
Length (cm)	10	30	33	40
Success Rate (%)	100	66.7	33.3	0
$\frac{\text{Max reaching distance}}{\text{Total body length}}$ (%)	38			

Heterodon nasicus: Reached a maximum distance of 31.9% of its body length. Success rates dropped sharply beyond 15 cm, verifying the trade-off of its fossorial adaptation.

Boa constrictor： Achieved a 43% horizontal reach. Although it has the smallest tail ratio, its massive muscle mass allows it to sustain relatively long reaches.

Boiga kraepelini: Reached a maximum of 38%. Surprisingly, its horizontal performance was not as exceptional as its vertical performance. This may be due to its thin body, which makes the friction weaker when reaching horizontally.

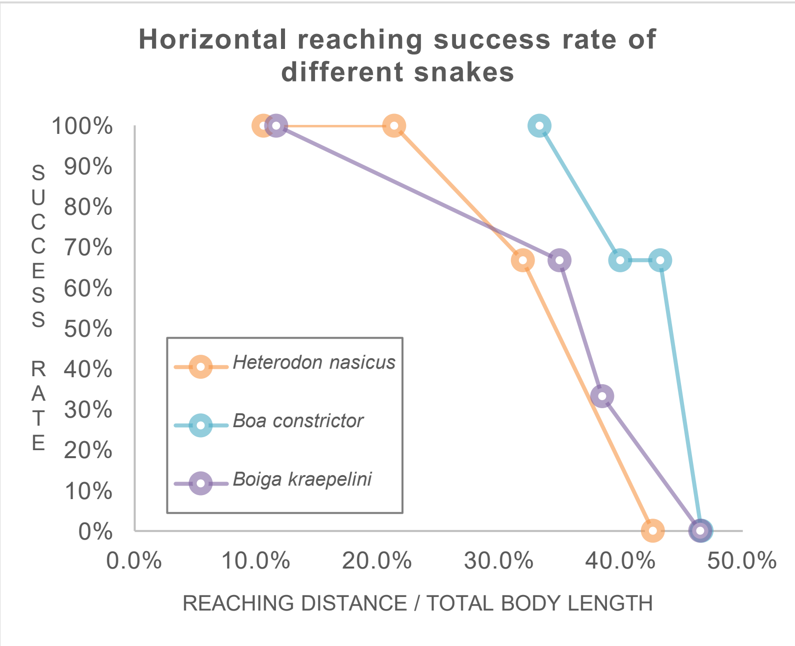
Heterodon nasicus：最大可達其身體長度的 31.9%。超過 15 公分後，成功率急劇下降，這驗證了其穴居適應性的權衡取捨。

Boa constrictor：水平伸展達到 43%。儘管它的尾部比例最小，但龐大的肌肉量使其能夠維持相對較長的伸展距離。

Boiga kraepelini：最大可達 38%。令人驚訝的是，其水平表現不如垂直表現出色。這可能是由於其體型纖細，導致水平伸展時摩擦力較弱。

Table 9: Three Species' Success Rate Comparison

	Horizontal reaching length (cm)										
Success rate (%)	5	10	15	20	30	33	40	50	60	65	70
Heterodon nasicus body length：47cm	100	100	66.7	0							
Boa constrictor body length：150cm								100	66.7	66.7	0
Boiga kraepelini body length：86cm		100			66.7	33.3	0				



Graph 4: Horizontal reaching success rate of different snakes

Table 9 shows the comparison of different snake's success rate in different reaching length together. Graph 4 indicates that the reaching limits for most snakes converge around 43% of their body length.

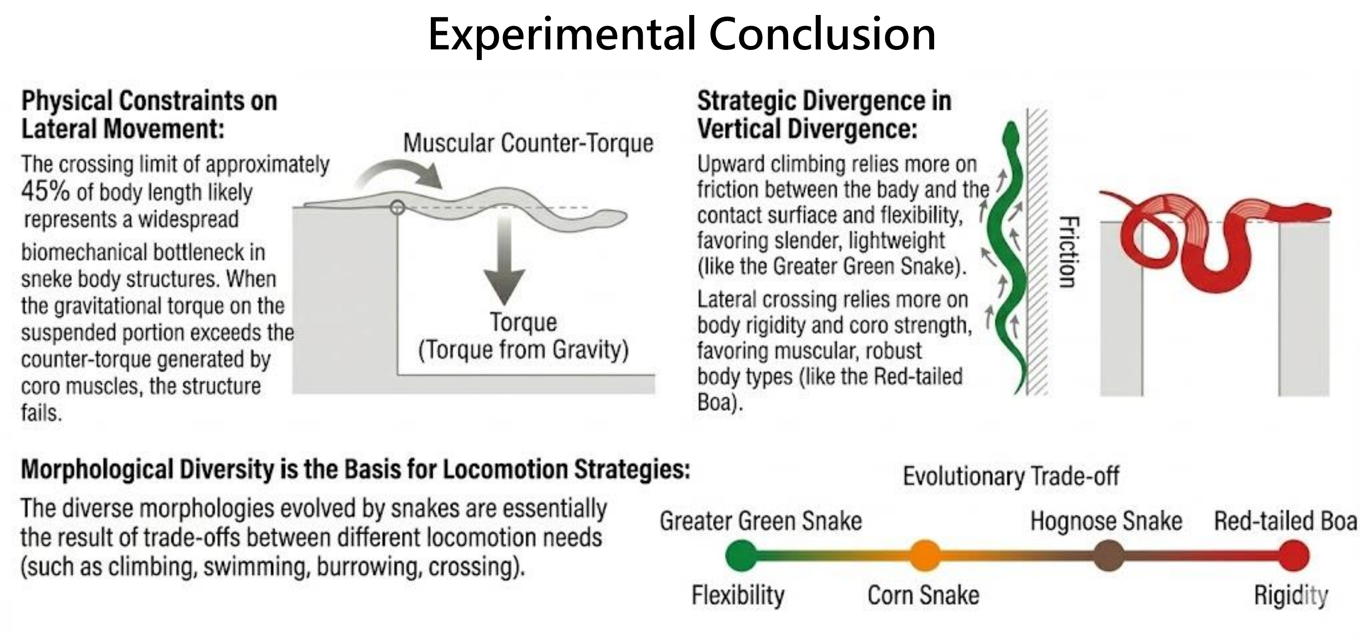
In graph 4, we can also see that the success rate of *Heterodon nasicus* dropped sharply unlike the other snakes.

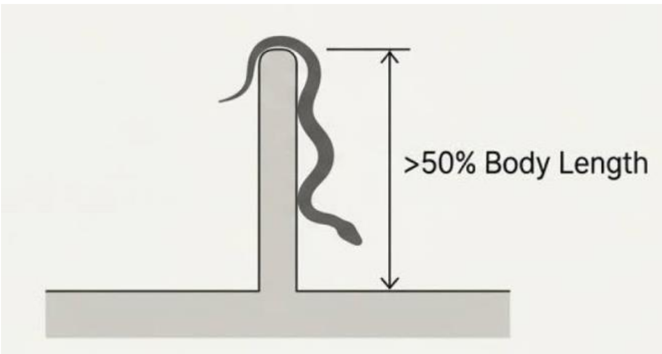
The "43% limit" is a biomechanical constraint for limbless animals. As the reaching distance increases, gravitational torque rises rapidly.

表 9 顯示了不同蛇類在不同伸展長度下的成功率比較。圖 4 表明，大多數蛇的伸展極限集中在其身體長度的 43%左右。

在圖 4 中，我們還可以看到，與其他蛇類不同，北美豬鼻蛇（Heterodon nasicus）的成功率急劇下降。

「43%極限」是無四肢動物的生物力學限制。隨著伸展距離的增加，重力扭矩會迅速上升。





A


B

Figure 5: From biomechanics to conservation: Engineering a safer coexistence. (AI-generated schematic diagram)

A: **Habitat Barriers:** Knowing the~47% limit allows us to design effective,non-harmful barriers. A barrier. exceeding 50% of a local snake's max body length can prevent entry intoresidential areas.

B: **Eco-Corridors** : Our findings support designing canopy bridges over roads for arboreal snakes like the Boiga, which are highly vulnerable on flat ground, to reduce roadkill.

Figure 6: The snake's tail as a'fifth limb': A blueprint for next-generation robotics. (AI-generated schematic diagram)

The Challenge: Current snake-like robots struggle with crossing gaps in discontinuous terrain.

The Solution: By mimicking the Boiga's high-tail-ratio structure as a posterior anchor and counterbalance, we can inspire robots that perform stable cantilevered extensions to navigate rubble and reach victims in disaster zones.